visStatistics: The right test, visualised

by Sabine Schilling

Abstract `visStatistics` automatically selects and visualises appropriate statistical hypothesis tests comparing two vectors of class `"numeric"`, `"integer"`, `"factor"`, or `"ordered"`. No manual test selection is required: the function determines the appropriate method from the variable classes, distributional assumptions, and sample sizes. The main function `visstat()` visualises the selected test with appropriate graphs, including box plots, bar charts, regression lines with confidence bands, mosaic plots, residual plots, and Q-Q plots. These outputs are annotated with relevant test statistics, assumption checks, and post-hoc analyses where applicable. The package shifts attention from ad-hoc test selection to visual diagnostic assessment and statistical interpretation. This is particularly relevant for routine frequentist tests, where users often select among common procedures without formally checking assumptions such as residual normality, variance homogeneity, sample size requirements, or expected cell counts. The scripted workflow is well suited for browser-based and server-side R applications, where users interact through a web interface while R handles the data processing. Other typical use cases include quick visualisations, statistical consulting, and teaching settings where transparent test selection and diagnostic feedback are important.

1 Introduction

In the frequentest tradition, most routine data analyses reduce to a comparatively small set of inferential frameworks, including group comparisons, regression models and contingency-table analyses (**?**, 2009,Hayat et al. (2017), Sato et al. (2017), Brodeur et al. (2020)); their correct use depends on assumptions that are often checked informally or not at all. `visStatistics` targets this gap by making routine frequentist test selection explicit, assumption-aware, visual, and reproducible. It selects out a test by using a reproducible decision framework based on the data's distributional assumptions and sample size. It visualises its decision by, where appropriate, assumption-diagnostic plots and a descriptive plot with the main test statistics annotated, and returns an R object whose `print()` and `summary()` methods expose the complete test results. The scripted workflow is well suited for browser-based applications where sensitive data (such as highly confidential medical records) is stored securely on a server and can not be directly accessed by users. This approach was already successfully applied to develop a medical scoring tool (Bijlenga et al., 2017). When selecting tests for group comparisons, packages with overlapping scope such as `automatedtests` (Zeevat, 2025) and `compareGroups` (Subirana et al., 2014) base their test selection on group-wise normality assessments of the response variable. But group-wise testing inflates the family-wise type I error rate: the probability of at least one false rejection of normality grows with the number of compared groups, leading to unnecessary switching to non-parametric methods even when the linear model assumptions hold globally. In contrast, `visStatistics` assesses the normality of the overall model residuals, adhering to the general linear model framework (Searle, 1971).

2 Package overview

visStatistics is available on CRAN, its main function 'vistat()" accepts input in two equivalent forms:

```
# Recommended form:
visstat(x, y)

# Formula interface:
visstat(y ~ x, data = dataframe)
```

where `x` and `y` are vectors of class `"numeric"`, `"integer"`, or `"factor"`. The function returns an object of class `"visstat"` with `print()`, `summary()`, and `plot()` methods.

3 Decision logic

The decision logic '`visstatistics()` summarised in figure @ref(fig:overview is driven by the class of its input vectors:

Numeric responses with categorical predictors enter the central-tendency branch; two unordered factors enter the proportion-comparison branch; two numeric variables enter simple linear regression

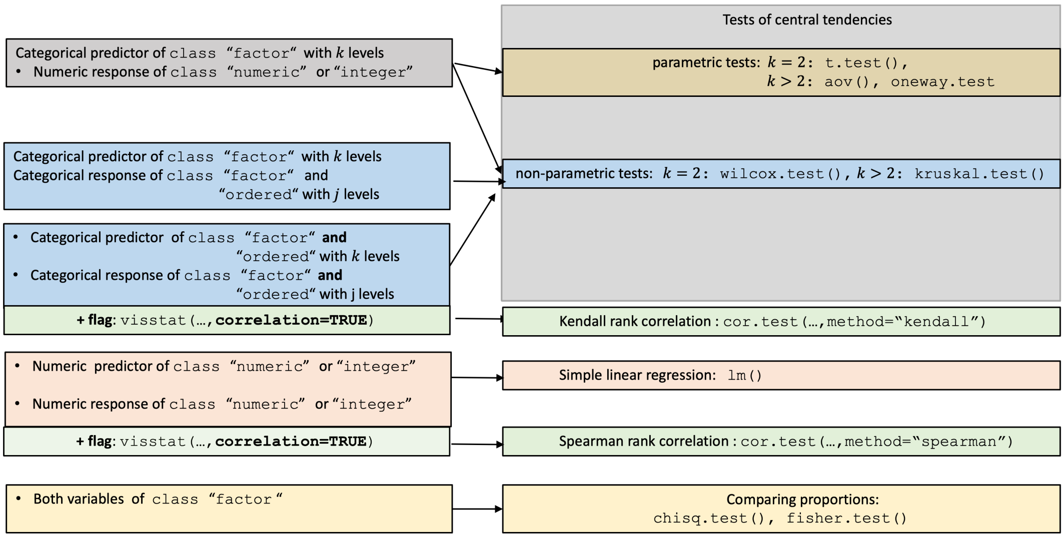

Figure 1: Overview of all implemented tests selected based on input class.

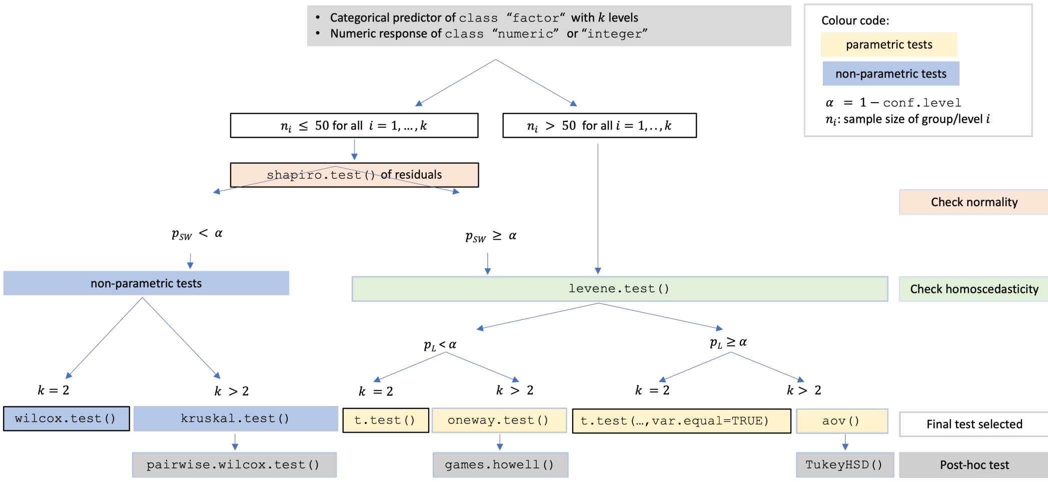

Figure 2: Decision tree for tests on central tendencies. If all groups contain more than 50 observations, the formal residual normality test is bypassed and variance homogeneity is assessed directly. Otherwise, Shapiro–Wilk on standardised residuals determines whether the route remains mean-based or switches to rank-based tests; Levene–Brown–Forsythe then selects equal-variance or Welch-type procedures.

unless `correlation = TRUE`, which switches to Spearman rank correlation; and two factors, which inherit the class `ordered` are analysed as Kendall rank correlation when `correlation = TRUE`. Definitions of all implemented test statistics and association measures are collected in Appendix B.

3.1 Test selection based on the assumptions of the general linear framework

Student's t-test, Fisher's ANOVA, and simple linear regression all belong to the general linear model framework see Appendix A and thus share a common set of assumptions: the expected value of the response is a linear function of the predictors, the error terms are independent and normally distributed with expectation 0, and the error variance is constant at σ^2. Test selection for tests of central tendencies is steered by checking the fitted model residuals in regard to these assumptions.

In the following subsections, we detail the decision logic following the structure of the overview figure 1.

3.2 Numeric response and categorical predictor: comparing central tendencies

Figure 2 expands the central-tendency branch.

The first split checks group size. When all groups contain more than 50 observations, the normality of the sampling distribution of the group means is assumed by the central limit theorem (Lumley et al., 2002; Rasch et al., 2011), and the residual normality test is bypassed to avoid rejecting negligible deviations in large samples (Ghasemi and Zahediasl, 2012; Fagerland, 2012; Shatz, 2024). Otherwise, a linear model `lm(y ~ x)` (see Appendix A) is fitted, assumption diagnostic plots are displayed (for details see Assumption diagnostics), and the Shapiro–Wilk (SW) test (Shapiro and Wilk, 1965) is applied to internally studentised residuals from `rstandard()` (Cook and Weisberg, 1982).

If normality is rejected ($p_{SW} \leq \alpha$), non-parametric tests are selected: `wilcox.test()` (Mann and Whitney, 1947) for two groups, or `kruskal.test()` (Kruskal and Wallis, 1952) followed by Holm-adjusted `pairwise.wilcox.test()` (Holm, 1979) for more than two groups.

If normality is not rejected, the Levene–Brown–Forsythe test `levene.test()` (Brown and Forsythe, 1974) (see Appendix B) assesses variance homogeneity. For homoscedastic data, `t.test(var.equal = TRUE)` is applied for two groups, or Fisher's `aov()` with `TukeyHSD()` for more than two groups. For heteroscedastic data, Welch's `t.test()` (Welch, 1947; Satterthwaite, 1946) is applied for two groups, or Welch's `oneway.test()` (Welch, 1951) with `games.howell()` (Games and Howell, 1976) for more than two groups. When switching to Welch methods, group-wise normality diagnostics are additionally displayed, since the pooled `lm()` residuals are no longer the appropriate diagnostic under heteroscedasticity.

3.3 Both variables numeric: regression or Spearman rank correlation

By default, `visstat()` fits a simple linear regression via `lm()`, regardless of whether GLM assumptions are met, and always displays assumption diagnostics. When `correlation = TRUE` is set explicitly, Spearman's ρ is computed via `cor.test(..., method = "spearman")`. Because the choice between modelling a directional relationship and measuring a monotone association cannot be automated from data types alone, Spearman correlation is never triggered automatically.

3.4 Both variables categorical: comparing proportions

When both variables are categorical, `visstat()` tests independence using Pearson's χ^2 test (`chisq.test()`) or Fisher's exact test (`fisher.test()`), depending on expected cell counts following Cochran's rule (Cochran, 1954): the χ^2 approximation is used if no expected cell count is less than 1 and no more than 20% of cells have expected counts below 5. Yates' continuity correction (Yates, 1934) is applied by default to 2×2 tables. For general $R \times C$ tables, Pearson's χ^2 result is supplemented with a mosaic plot from **vcd** (Meyer et al., 2006, 2024) with tiles coloured by Pearson residuals.

3.5 Ordered factors

When the response is an ordered factor, `visstat()` converts it to integer level codes and follows the non-parametric path. When both variables are ordered and `correlation = TRUE`, Kendall's τ_b (Kendall, 1945; Agresti, 2010) is computed instead, as it corrects for ties explicitly — unavoidable with few ordinal levels (Xu et al., 2013).

4 Examples

The examples follow the same order as the decision logic. The number of plots returned by `visstat()` depends on the input types and the outcome of assumption checks. Most tests produce two panels: an assumption-diagnostic panel and a result panel. When heteroscedasticity is detected and `visstat()` routes to a Welch method, a third panel with group-wise normality diagnostics is added, since pooled-residual diagnostics are no longer appropriate under unequal variances. Tests that have no separate assumption diagnostics, such as Kendall's τ_b for two ordered factors, produce a single plot.

The datasets were chosen to exercise the main branches of the decision tree: normal and homoscedastic residuals leading to Student's t-test or one-way ANOVA; normal residuals with unequal variances leading to Welch's t-test or Welch's one-way test; non-normal residuals leading to Wilcoxon or Kruskal–Wallis tests, with Holm-adjusted pairwise Wilcoxon tests for the multi-group extension; numeric–numeric regression and rank correlation; categorical independence tests under and beyond Cochran's rule; and ordinal rank correlation.

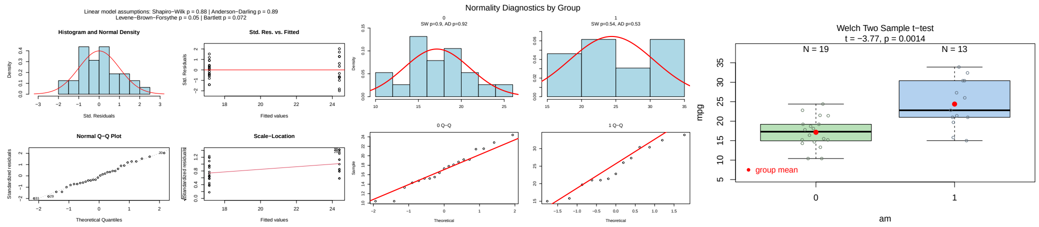

Figure 3: Welch's t-test applied to the `mtcars` dataset (`am` vs. `mpg`). Pooled-residual assumption diagnostics (Shapiro–Wilk, Bartlett, Levene), group-wise normality diagnostics, and box plots of fuel efficiency by transmission type with the Welch t-test result.

4.1 Numeric response and categorical predictor: two groups

For two groups, `visstat()` first evaluates the linear-model assumptions on pooled residuals. If Shapiro–Wilk rejects normality, the decision path switches to the Wilcoxon rank-sum test. If normality is not rejected, the Levene–Brown–Forsythe test decides between Student's equal-variance t-test and Welch's unequal-variance t-test.

Welch's t-test

The *Motor Trend Car Road Tests* dataset (`mtcars`) contains 32 observations, where `mpg` denotes miles per (US) gallon and `am` represents the transmission type (`0` = automatic, `1` = manual). With binary factor `am` and continuous response `mpg`, Shapiro–Wilk does not reject normality of the pooled residuals, while the Levene–Brown–Forsythe test detects heteroscedasticity. The routing therefore leads to Welch's t-test rather than Student's t-test.

```
mtcars$am <- as.factor(mtcars$am)
t_test_stats <- visstat(mtcars$am, mtcars$mpg)
```

Wilcoxon rank-sum test

The Wilcoxon branch is exemplified on examination grades for a class of 44 students (21 girls, 23 boys). After fitting `lm(grade ~ sex)`, Shapiro–Wilk rejects normality of the pooled residuals, so `visstat()` switches to the non-parametric Wilcoxon rank-sum test.

```
grades_gender <- data.frame(
  sex   = as.factor(c(rep("girl", 21), rep("boy", 23))),
  grade = c(19.3, 18.1, 15.2, 18.3,  7.9,  6.2, 19.4, 20.3,  9.3, 11.3,
            18.2, 17.5, 10.2, 20.1, 13.3, 17.2, 15.1, 16.2, 17.0, 16.5,
             5.1, 15.3, 17.1, 14.8, 15.4, 14.4,  7.5, 15.5,  6.0, 17.4,
             7.3, 14.3, 13.5,  8.0, 19.5, 13.4, 17.9, 17.7, 16.4, 15.6,
            17.3, 19.9,  4.4,  2.1)
)
wilcoxon_stats <- visstat(grades_gender$sex, grades_gender$grade)
```

4.2 Numeric response and categorical predictor: more than two groups

For more than two groups, the same residual-normality decision is applied first. If residual normality is rejected, `visstat()` applies `kruskal.test()`. Otherwise, `levene.test()` chooses between Fisher's one-way ANOVA (`aov()`) and Welch's heteroscedastic one-way ANOVA (`oneway.test()`). The graphical output consists of two panels. The first panel displays standardised residuals versus fitted values and a normal Q–Q plot; no more than about 5% of the standardised residuals should exceed $|2|$ when residuals are normally distributed, and the Q–Q points should approximately follow the red reference line. The title reports the Shapiro–Wilk and Anderson–Darling normality tests together with Bartlett's and Levene–Brown–Forsythe tests for homoscedasticity. The second panel shows the selected omnibus test as box plots with sample sizes above each group; for the parametric branches, the title also reports the corresponding F statistic. When the largest group has no more than 50 observations, jittered points are overlaid to show the raw data.

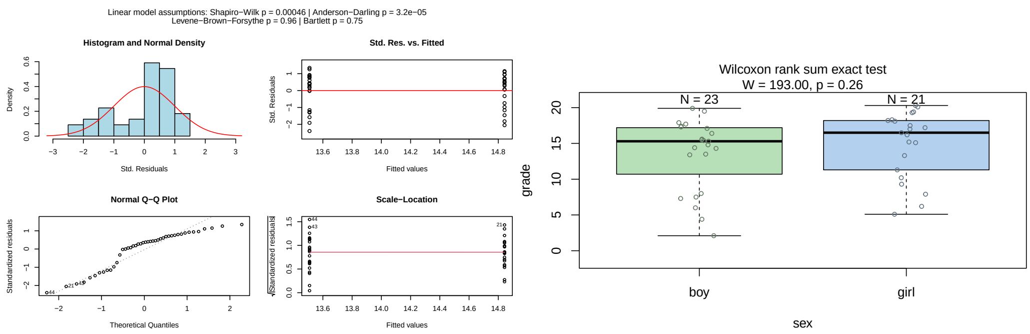

Figure 4: Wilcoxon rank-sum test applied to examination grades by sex. Assumption diagnostics (Shapiro–Wilk rejected; non-parametric path selected) and box plots with the Wilcoxon test result.

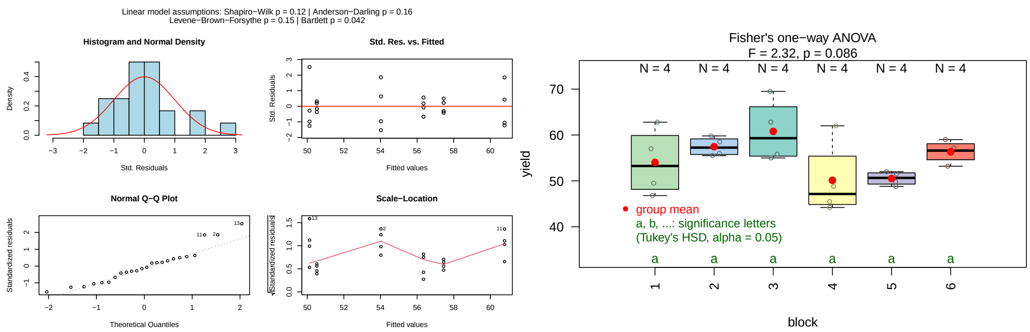

Figure 5: Fisher's one-way ANOVA applied to the `npk` dataset (`block` vs. `yield`). Assumption diagnostics and box plots of pea yield by experimental block; compact letter display (TukeyHSD, $\alpha = 0.05$) shows no significant pairwise differences.

ANOVA, Welch ANOVA, and Kruskal–Wallis are omnibus tests: a significant result tells us that *some* group differs, but not which. To identify the differing pairs, `visstat()` tests all pairwise comparisons among the factor levels, defining a family of tests (Abdi, 2007). Because the three omnibus tests rest on different assumptions, each branch uses a matching post-hoc procedure: `TukeyHSD()` after `aov()`, `games.howell()` after `oneway.test()`, and `pairwise.wilcox.test(p.adjust.method = "holm")` after `kruskal.test()`. `TukeyHSD()` controls the family-wise error rate through the studentised range distribution under a common-variance assumption. `games.howell()` uses separate variance estimates and Welch-adjusted degrees of freedom for each pair, making it the post-hoc procedure for the heteroscedastic Welch branch. The Kruskal–Wallis branch uses Holm's step-down adjustment for the pairwise Wilcoxon tests. Pairs whose adjusted post-hoc p-value falls below α are marked with different green letters below the box plots; pairs sharing a letter are not significantly different. The returned object contains the corresponding adjusted p-values.

One-way ANOVA with Tukey HSD

The `npk` dataset reports the yield of peas from an agricultural experiment conducted on six blocks. Shapiro–Wilk does not reject normality of the pooled residuals and the Levene test does not reject homoscedasticity, so `visstat()` applies Fisher's one-way ANOVA followed by Tukey HSD post-hoc comparisons. The omnibus F-test is not significant at $\alpha = 0.05$, and all groups share the same compact-letter label.

```
anova_npk <- visstat(npk$block, npk$yield, conf.level = 0.95)
```

Kruskal-Wallis rank sum test

The `iris` dataset contains petal-width measurements for three species. Here, the residual plots and very small Shapiro–Wilk and Anderson–Darling p-values indicate non-normal residuals. Since Shapiro–Wilk falls below α, `visstat()` switches to `kruskal.test()` followed by Holm-adjusted `pairwise.wilcox.test()`. All three species differ significantly in petal width, as indicated by distinct compact-letter labels.

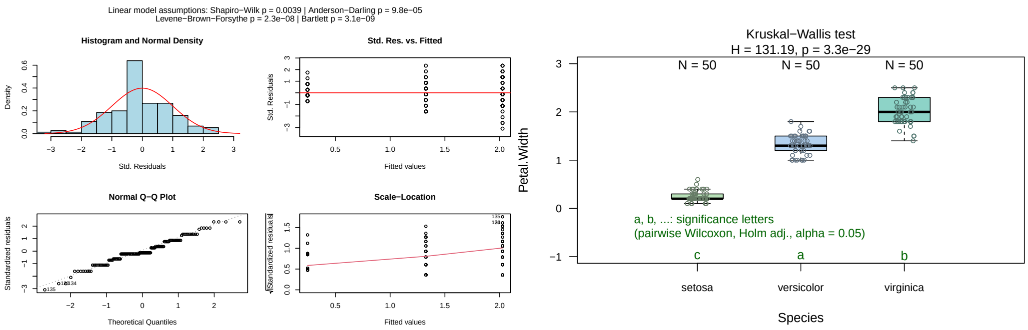

Figure 6: Kruskal-Wallis test applied to the `iris` dataset (`Species` vs. `Petal.Width`). Assumption diagnostics (Shapiro–Wilk rejected) and box plots with the Kruskal-Wallis result; all pairwise post-hoc comparisons significant (all letters differ).

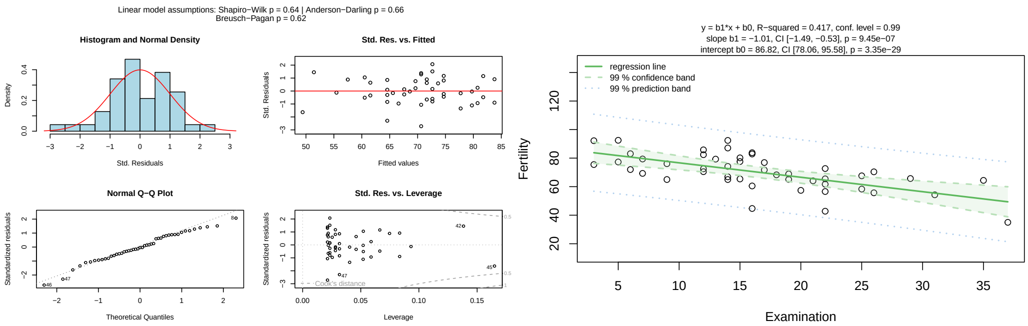

Figure 7: Simple linear regression of `Fertility` on `Examination` for the `swiss` dataset (`conf.level = 0.99`). Assumption diagnostics (Shapiro–Wilk, Anderson–Darling, Breusch–Pagan) and scatter plot with fitted regression line, 99% confidence band (dark shading), and 99% prediction band (light shading).

```
kruskal_iris <- visstat(iris$Species, iris$Petal.Width)
```

4.3 Both variables numeric

Simple linear regression

`When both predictor and response are numerical, visstat()` checks the standardised residuals from `lm()` graphically and with the Shapiro–Wilk and Anderson–Darling tests; the Breusch–Pagan test assesses homoscedasticity. Regardless of these diagnostics, a simple linear regression fitted and displayed. The graphical shows the fitted line $\hat{y} = b_0 + b_1 x$, where b_0 is the intercept and b_1 the slope, with pointwise confidence and prediction bands at the specified `conf.level`.

The title shows R^2, the estimated regression parameters with confidence intervals and p-values. The returned object contains the regression statistics, the residual-normality p-values, and pointwise estimates of the confidence and prediction bands.

The `swiss` dataset records standardised fertility and socioeconomic indicators for 47 French-speaking Swiss provinces in 1888. We examine how the share of draftees achieving the highest army examination score (`Examination`) predicts the fertility measure (`Fertility`), with `conf.level = 0.99`. Both normality tests pass and the Breusch–Pagan test confirms homoscedasticity, validating the linear model.

```
linreg_swiss <- visstat(swiss$Examination, swiss$Fertility, conf.level = 0.99)
```

Spearman rank correlation

For numeric–numeric input, correlation analysis is never triggered automatically; it requires the explicit flag `correlation = TRUE`. With wind speed (`Wind`) and ozone concentration (`Ozone`) from the `airquality` dataset (Chambers, 2018), a default `visstat()` call fits simple linear regression but

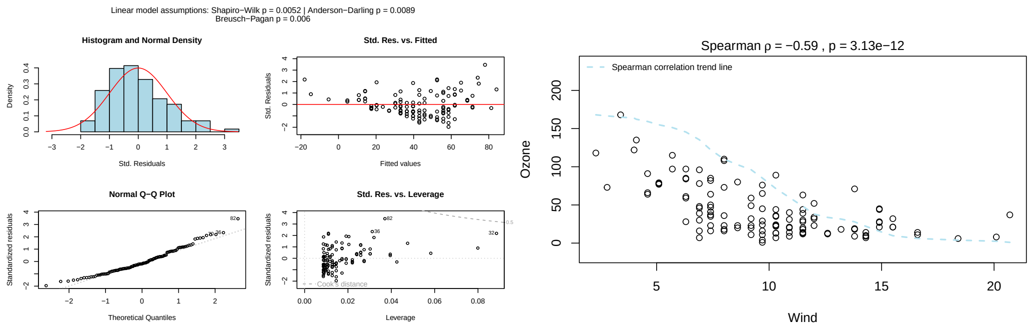

Figure 8: Spearman rank correlation of `Wind` and `Ozone` from the `airquality` dataset (`correlation = TRUE`). Assumption diagnostics and scatter plot of ranked values annotated with ρ and the p-value.

flags violated normality and homoscedasticity. The warning recommends two paths: rerun with `correlation = TRUE` within `visstat()`, or switch to a model outside it. Staying within `visstat()`, `correlation = TRUE` gives an assumption-free Spearman result at the cost of causal interpretability (Note that here e.g. a Gamma model with log link might be more appropriate to describe the data, as `Ozone` is strictly positive and continuous, and its variance grows with the mean),

```
spearman_air <- visstat(airquality$Wind, airquality$Ozone, correlation = TRUE)
```

4.4 Both variables categorical

Observed frequencies are arranged in a contingency table, where rows index the levels of the response and columns index the levels of the predictor. `visstat()` tests the null hypothesis that the variables are independent. If expected frequencies satisfy Cochran's rule, Pearson's χ^2 test is used; otherwise, `visstat()` switches to Fisher's exact test. For 2×2 tables, Yates' continuity correction is applied by default to Pearson's χ^2 test. The graphical output depends on the selected test. General $R \times C$ Pearson χ^2 tests show a grouped column plot of row percentages with the p-value in the title, followed by a mosaic plot with colour-coded Pearson residuals. Yates-corrected 2×2 χ^2 tests show the grouped column plot only, because the corrected statistic is not decomposed into cell-level residuals. Fisher's exact test shows absolute counts with N labels above each bar and the p-value in the title.

The following examples are based on the `HairEyeColor` contingency table, which is converted to the column-based data frame expected by `visstat()` using `counts_to_cases()`.

Pearson's χ^2 test

With `Eye` and `Hair` from `HairEyeColor`, all expected cell counts exceed the Cochran thresholds (Cochran, 1954), so the χ^2 approximation is used. For this 4×4 table, `visstat()` returns both a grouped bar chart of row percentages and a mosaic plot with tiles coloured by Pearson residuals. The mosaic plot shows which cells drive the association: blue tiles indicate observed counts above expectation and red tiles indicate observed counts below expectation.

```
hair_eye_df <- counts_to_cases(as.data.frame(HairEyeColor))
visstat(hair_eye_df$Eye, hair_eye_df$Hair)
```

Fisher's exact test

Restricting `HairEyeColor` to male participants with black or brown hair and hazel or green eyes yields a 2×2 table where one expected frequency is less than 5, violating Cochran's rule (Cochran, 1954). `visstat()` therefore applies Fisher's exact test. For 2×2 tables, the returned object also contains the conditional maximum likelihood estimate of the odds ratio and its confidence interval.

```
hair_eye_male <- HairEyeColor[, , 1]
black_brown_hazel_green <- hair_eye_male[1:2, 3:4]
black_brown_hazel_green_df <- counts_to_cases(
  as.data.frame(black_brown_hazel_green))
```

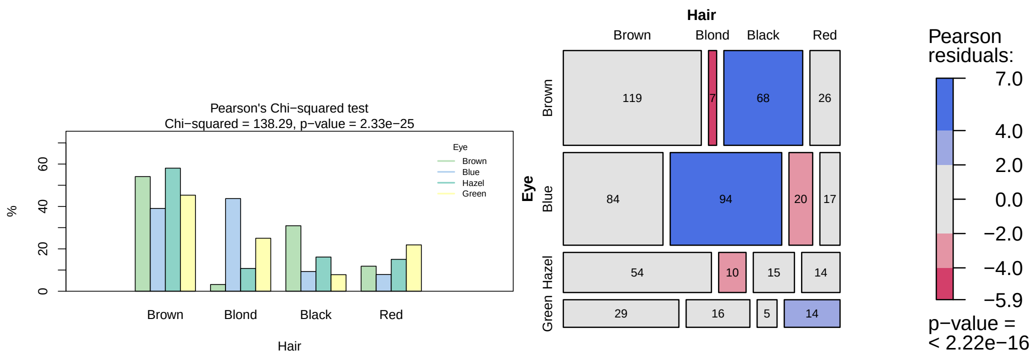

Figure 9: Pearson's χ^2 test applied to the `HairEyeColor` dataset. Grouped bar chart of eye colour by hair colour and mosaic plot with tiles coloured by Pearson residuals (blue: over-represented, red: under-represented).

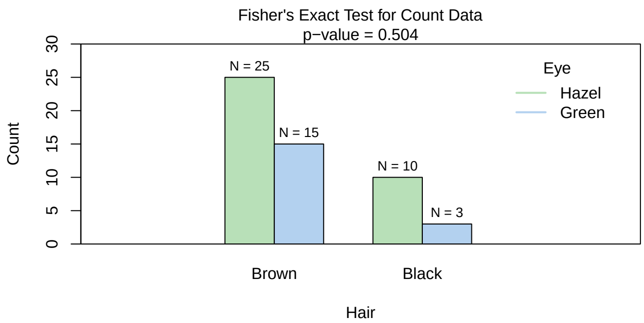

Figure 10: Fisher's exact test applied to a 2×2 subset of `HairEyeColor` (male participants, black/brown hair, hazel/green eyes). Bar chart of absolute counts with the p-value; the odds ratio and confidence interval are accessible from the returned object.

```
fisher_stats <- visstat(black_brown_hazel_green_df$Eye,
                        black_brown_hazel_green_df$Hair)
fisher_stats$estimate   # odds ratio
```

```
#> odds ratio
#>  0.5062015
```

```
fisher_stats$conf.int   # 95 % CI
```

```
#> [1] 0.07725895 2.40885255
#> attr(,"conf.level")
#> [1] 0.95
```

4.5 Ordered factors

When the response is ordered and the predictor is categorical, `visstat()` converts the response to integer level codes and redirects to the non-parametric Wilcoxon or Kruskal–Wallis path above. When both variables are ordered factors and `correlation = TRUE`, `visstat()` instead tests for a monotone association via Kendall's τ_b (Kendall, 1945; Agresti, 2010). The Kendall output is a single jittered rank–rank scatter plot that visualises the monotone trend, colour-codes points by the predictor level, and annotates the plot with τ_b and the p-value.

Kendall's τ_b

A hypothetical survey of 150 secondary-school students records alcohol consumption frequency and academic performance on five-point ordinal scales. A negative monotone association is induced by construction: students who consume alcohol more frequently tend to have lower academic performance.

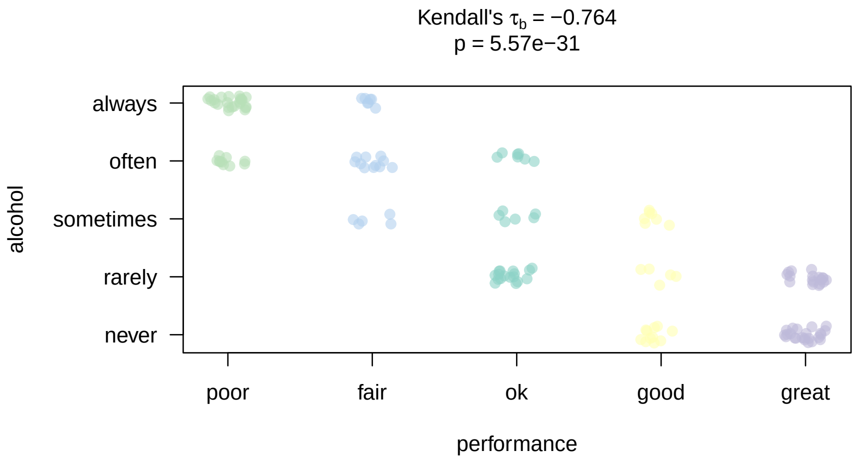

Figure 11: Kendall's τ_b for a hypothetical survey ($n = 150$): alcohol consumption frequency vs. academic performance (both five-point ordinal scales). Jittered rank–rank scatter annotated with τ_b and the p-value; the negative monotone association is significant.

```
set.seed(42)
n <- 150
xs <- sample(1:5, n, replace = TRUE)
ys <- pmin(5, pmax(1, (6 - xs) + sample(-1:1, n, replace = TRUE)))
likert_alc  <- c("never", "rarely", "sometimes", "often", "always")
likert_perf <- c("poor",  "fair",   "ok",        "good",  "great")
alcohol     <- ordered(likert_alc[xs],  levels = likert_alc)
performance <- ordered(likert_perf[ys], levels = likert_perf)
kendall_result <- visstat(performance, alcohol, correlation = TRUE)
```

5 Limitations

The main purpose of this package is a decision-logic based automatic selection visualisation of the "right" statistical test. Therefore, except for the user-adjustable `conf.level` parameter, all statistical tests are applied using their default settings from the corresponding base R functions. As a consequence, paired tests are currently not supported and `visstat()` does not allow to study interactions terms between the different levels of an independent variable in an analysis of variance. Focusing on the graphical representation of tests, only simple linear regression is implemented, as multiple linear regressions cannot be visualised. When regression assumptions are violated, the package offers Spearman rank correlation (`correlation = TRUE`) as the sole alternative to linear regression. Alternative methods such as data transformation, generalised linear models or robust regression are not implemented: each requires user judgment – about the transformation family, the link function, or the estimator – that cannot be automated without substantially expanding the decision tree and increasing the risk of Type I error inflation.

Within this restricted automated pipeline, the routing depends on p-values of assumption tests to select tests of central tendencies. But no single (assumption) test maintains optimal Type I error rates and statistical power across all distributions (Olejnik and Algina, 1987), and p-values obtained from these tests may be unreliable if their assumptions are violated.

Assessing assumptions solely through p-values can therefore lead to both type I errors (false positives) and type II errors (false negatives). In large samples, even minor, random deviations from the null hypothesis can result in statistically significant p-values, leading to type I errors. Conversely, in small samples, substantial violations of the assumption may not reach statistical significance, resulting in type II errors (Kozak and Piepho, 2018).

Moreover, assumption tests provide no information on the nature of deviations from the expected distribution (Shatz, 2024). Thus the assessment of normality or homoscedasticity should never rely solely on p-values but should be complemented by visual inspection of the diagnostic plots generated by `visstat()`.

This general limitation is particularly relevant for normality test. Normality tests behave poorly at both ends of the sample-size range: with small samples they fail to detect non-normality, and with large samples they flag negligible departures from normality as significant (Ghasemi and Zahediasl, 2012; Fagerland, 2012; Franc, 2025).

For this reason, this package uses $n > 50$ per group as a rule of thumb threshold to omit normality testing, assuming parametric tests of central tendencies to become robust to non-normality at larger sample sizes based on the central limit theorem. The threshold is necessarily arbitrary; simulation studies show that the convergence rate depends on the skewness and kurtosis of the underlying distribution, with moderately skewed distributions requiring roughly 40–50 observations for adequate convergence of the sampling distribution of the mean (Fagerland, 2012). Simulation studies suggest that Shapiro–Wilk has the highest power among normality tests in small to moderate ($n = 10$ to 100) sample sizes (Razali and Wah, 2011). Accordingly, Shapiro–Wilk is used to select between parametric (ANOVA/Welch's t-test) and non-parametric (Kruskal–Wallis/Wilcoxon) methods for groups smaller than 50.

The same issue extends to combinations of assumption tests. Combining tests for normality and homoscedasticity using simple majority voting inflates the overall Type I error rate. Therefore, automated test selection based solely on p-values cannot replace the visual inspection of sample distributions provided by `visstat()`. Based on the provided diagnostic plots, it may be necessary to override the automated choice of test in individual cases.

To avoid further compounding these problems, the automated test selection pipeline deliberately restricts its scope to the most frequently used tests, particularly those relevant in a medical context (Strasak et al., 2007, Sato et al. (2017),Chicco et al. (2025)). While one of R's greatest strengths is the sheer volume of statistical methods available, incorporating a wider array of methods would require additional preliminary assumption checks, which in turn exacerbates the risk of overall Type I error inflation. Furthermore, expanding the pipeline would result in a highly complex decision tree, rendering the underlying statistical logic increasingly opaque to the user.

The package depends only on base R graphics with no `ggplot2` (Wickham, 2016) dependency, keeping the transitive dependency footprint minimal. For more polished, annotated plots of chosen statistical tests, we refer to packages such as `ggstatsplot` (Patil, 2021) or `ggpubr` (Kassambara, 2026).

Bootstrapping methods (Wilcox, 2021) make minimal distributional assumptions and can provide confidence intervals for nearly any statistic. However, bootstrapping is computationally intensive, often requiring thousands of resamples, and may perform poorly with very small sample sizes. The computational intensity of bootstrap runs counter to the purpose of the `visStatistics` package, which is designed to offer a rapid overview of the data, laying the groundwork for deeper analysis in subsequent steps.

6 Summary

visStatistics provides a single-function interface to automated statistical test selection, execution, and visualisation for the most commonly used inferential frameworks in two-variable analyses. Its principal contribution is the integration of the full pipeline — data-type detection, distributional assumption checking on GLM residuals, test routing, and annotated graphical output — into a reproducible, server-deployable workflow with a minimal dependency footprint. The package is available on CRAN at `https://CRAN.R-project.org/package=visStatistics` and on GitHub at `https://github.com/shhschilling/visStatistics`.

7 Appendix A: The general linear model

The general linear model (Searle, 1971) provides a unified mathematical framework underlying Student's t-test, Fisher's ANOVA, and simple linear regression.

Let n denote the number of observations and $k-1$ the number of predictors. The model for observation i, $i = 1, \ldots, n$ is:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{k-1} x_{i,k-1} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$$

where Y_i is the response for observation i, x_{ij} is the value of predictor j for observation i, $\beta_0, \beta_1, \ldots, \beta_{k-1}$ are the k parameters, and ε_i are independently distributed error terms with constant variance σ^2.

Student's t-test uses one binary indicator variable x_{i1}; testing $H_0 : \beta_1 = 0$ is equivalent to testing $H_0 : \mu_1 = \mu_2$.

Fisher's ANOVA uses $k-1$ indicator variables for k groups; testing $H_0 : \beta_1 = \cdots = \beta_{k-1} = 0$ is equivalent to testing equality of all group means.

Simple linear regression uses one continuous predictor; testing $H_0 : \beta_1 = 0$ examines whether a linear relationship exists.

In the two-sample case, the squared test test statistic of the Student's t-tests equals the Fisher's ANOVA test static, $t^2 = F$, resulting in identical p-values for `t.test(var.equal = TRUE)` and `aov()` .

8 Appendix B: The general linear model assumption diagnostics `vis_lm_assumptions()`

All tests within the linear model framework (see Appendix A) share a common set of assumptions:

- Linearity: The expected value of the response is a linear function of the predictors; assessed by checking for systematic patterns in residual plots.
- Error terms are independent, normally distributed with expectation value 0 and have constant error variance σ^2

The function `vis_lm_assumptions()` provides a unified visual and statistical framework for validating the [linear model] (#glm) assumptions. It fits a linear model using `aov()` and provides four standard diagnostic plots:

1. **Histogram and Normal Density**: Displays the distribution of standardised residuals with a red normal density curve overlay to visually assess normality.
2. **Std. Residuals vs. Fitted**: It shows the standardised residuals against fitted values with a zero-line to detect non-linearity or heteroscedasticity.
3. **Normal Q-Q Plot**: A theoretical quantile-quantile plot using the standardised residuals to identify deviations from the Gaussian distribution.
4. If `correlation = FALSE` (regression mode), the **Standardised Residuals vs. Leverage** plot; if `correlation = TRUE` (correlation mode), a **Scale-Location** plot.

8.1 Test for normality and homoscedasticity

The diagnostic plots are enhanced by p-values of tests for normality and homoscedasticity, shown in the title of the plot.

Normality Assessment: The function evaluates residual normality using both the Shapiro–Wilk test (`shapiro.test()`) and the Anderson–Darling test (`ad.test()`) (Gross and Ligges, 2015). Anderson–Darling is particularly sensitive to tail deviations (Razali and Wah, 2011; Yap and Sim, 2011).

Homoscedasticity Assessment: Variance equality is tested using the Levene-Brown-Forsythe test (`levene.test()`) (Brown and Forsythe, 1974) and Bartlett's test (`bartlett.test()`) in the ANOVA path.

These tests compare the spread of data across different factor levels.

For simple linear regression, the Breusch-Pagan test (`bp.test()`) is implemented.

`bartlett.test()` and 'bp.test() have standalone implementations in the package to avoid dependencies on external packages (see Appendix C).

9 Appendix C: Test statistics and association measures

9.1 Normality tests

The package uses both the Shapiro-Wilk test as well ass the Anderson-Darling test to check the normality of the studentised residuals $r_i^* = e_i/(s\sqrt{1-h_{ii}})$, where e_i are the raw residuals, s^2 is the residual mean square, and h_{ii} is the leverage of observation i (Cook and Weisberg, 1982).

Shapiro–Wilk test `shapiro.test()`

The Shapiro–Wilk test (Shapiro and Wilk, 1965) evaluates whether a sample $x_1, \ldots, x_n$ comes from a normal distribution. Let $x_{(1)} \leq \cdots \leq x_{(n)}$ be its order statistics. Introduce a reference sample $Y_1, \ldots, Y_n$ of independent standard normal random variables, i.e. $Y_i \sim N(0,1)$ for all i, and let $Y_{(1)} \leq \cdots \leq Y_{(n)}$ be their order statistics used to construct the Shapiro–Wilk weights.

Let $m_i = \mathrm{E}(Y_{(i)})$ and $v_{ij} = \mathrm{Cov}(Y_{(i)}, Y_{(j)})$ for $i,j = 1, \ldots, n$. Define $\mathbf{m} = (m_1, \ldots, m_n)^\top$ and $V = (v_{ij})_{i,j=1}^n$.

The vector **m** contains the expected standard-normal order statistics, and V is their covariance matrix. Let $\mathbf{a} = (a_1, \ldots, a_n)^\top$ be the resulting vector of normalised weights for the ordered observed sample values

$$\mathbf{a} = \frac{V^{-1}\mathbf{m}}{\sqrt{(\mathbf{m}^\top V^{-1} V^{-1}\mathbf{m})}}.$$

Then the Shapiro–Wilk statistic is

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}.$$

W takes values in $(0, 1]$; values close to 1 indicate normality. Simulation studies show that the Shapiro–Wilk test has the highest power among normality tests for small to moderate sample sizes ($n = 10$ to 100) (Razali and Wah, 2011). It drives the parametric/non-parametric branching decision in `visstat()`; for groups with more than 50 observations normality is assumed by the central limit theorem (Lumley et al., 2002; Rasch et al., 2011).

The implementation uses `shapiro.test()` from **stats** (**?**).

Anderson–Darling test `ad.test()`

The Anderson–Darling test (**?**) is particularly sensitive to deviations in the tails of the distribution (Razali and Wah, 2011; Yap and Sim, 2011). Let $z_i = (x_{(i)} - \bar{x})/s,\ i = 1, 2, \ldots, n$ be the standardised order statistics of x_i, where s is the sample standard deviation, and let Φ denote the standard normal cumulative distribution function. The test statistic is

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^{n} (2i - 1) \left[\ln \Phi(z_i) + \ln(1 - \Phi(z_{n+1-i}))\right].$$

The implementation uses `ad.test()` from **nortest** (Gross and Ligges, 2015). Albeit the Anderson–Darling result is shown in all assumption plots, it does **not** drive the branching decision; only Shapiro–Wilk does.

9.2 Homoscedasticity tests

The Levene–Brown–Forsythe test `levene.test()`

The test (Brown and Forsythe, 1974) improves upon Levene's original proposal (Levene, 1960) by centring on the group median rather than the group mean, making it robust to skewed data and outliers (Allingham and Rayner, 2012). The implementation `levene.test()` mimics the default behaviour of `leveneTest()` in **car** (Fox and Weisberg, 2019) and is the sole driver of the parametric/Welch branching decision.

For each observation y_{ij} in group i, the absolute deviation from the group median $\tilde{y}_i$ is

$$z_{ij} = |y_{ij} - \tilde{y}_i|.$$

The test statistic is the F-statistic from a one-way ANOVA on the z_{ij} values:

$$F = \frac{(N-k) \sum_{i=1}^{k} n_i (\bar{z}_i - \bar{z})^2}{(k-1) \sum_{i=1}^{k} \sum_{j=1}^{n_i} (z_{ij} - \bar{z}_i)^2},$$

where k is the number of groups, $N = \sum_{i=1}^{k} n_i$ is the total sample size, n_i is the sample size of group i, $\bar{z}_i$ is the mean of the absolute deviations within group i, and $\bar{z}$ is the overall mean of all absolute deviations. Under the null hypothesis of equal variances the statistic follows $F(k-1, N-k)$.

Bartlett's test `bartlett.test()`

Bartlett's test (Bartlett, 1937) has greater power than the Brown–Forsythe test when normality holds (Allingham and Rayner, 2012). Its test statistic is

$$K^2 = \frac{(N-k)\ln s_p^2 - \sum_{i=1}^{k}(n_i - 1)\ln s_i^2}{1 + \frac{1}{3(k-1)}\left(\sum_{i=1}^{k}\frac{1}{n_i - 1} - \frac{1}{N-k}\right)},$$

where k is the number of groups, $N = \sum_{i=1}^{k} n_i$ is the total sample size, n_i is the sample size of group i, s_i^2 is the sample variance of group i, and s_p^2 is the pooled variance

$$s_p^2 = \frac{1}{N-k}\sum_{i=1}^{k}(n_i - 1)\, s_i^2.$$

Under the null hypothesis the statistic approximately follows $\chi^2(k-1)$. Bartlett's result is shown in the assumption plot but does **not** drive the branching decision; only `levene.test()` does.

Breusch–Pagan test `bp.test()`

For simple linear regression, group-based variance tests are not applicable. The Breusch–Pagan test (Breusch and Pagan, 1979) assesses whether the variance of the residuals from the regression model depends on the predictor values. Let e_i be the residuals from the fitted regression model and let $\hat{\sigma}^2 = n^{-1}\sum_{i=1}^{n} e_i^2$ denote the mean of the squred residuals. The auxiliary regression models $e_i^2/\hat{\sigma}^2$ as a function of the predictors. Let R^2_{aux} denote the coefficient of determination from this auxiliary regression. Crucially, while a high R^2 is typically a sign of a strong model, here it indicates a failure of the constant-variance assumption: a higher R^2_{aux} means the error variance follows a systematic, non-random pattern. The Breusch–Pagan statistic is then calculated as:

$$BP = nR^2_{\text{aux}},$$

which is compared asymptotically to a χ^2 distribution with degrees of freedom equal to the number of predictors in the auxiliary regression.

9.3 Student's t-test and Welch's t-test

The test statistic for Student's t-test (`t.test(var.equal = TRUE)`) is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where $\bar{x}_1$ and $\bar{x}_2$ are the sample means, n_1 and n_2 the sample sizes, and s_p the pooled standard deviation

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}},$$

with s_1^2 and s_2^2 the sample variances. The statistic follows a t-distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom.

Welch's t-test relaxes the homoscedasticity assumption. Its statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}},$$

with degrees of freedom approximated by the Welch–Satterthwaite equation (Welch, 1947; Satterthwaite, 1946):

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

Welch's methods outperform their classical counterparts when variances differ (Moser and Stevens, 1992; Fagerland and Sandvik, 2009; Delacre et al., 2017).

9.4 Wilcoxon rank-sum test

The two-level factor `x` defines two groups with sample sizes n_1 and n_2. All $N = n_1 + n_2$ observations are pooled and assigned ranks 1 to N. Let $W_1 = \sum_{i=1}^{n_1} R(x_{1,i})$ denote the rank sum of the first group, where $R(x_{1,i})$ is the rank of observation $x_{1,i}$ in the pooled sample. The test statistic returned by `wilcox.test()` (Mann and Whitney, 1947) is the Mann–Whitney U statistic of the first group:

$$W = U_1 = W_1 - \frac{n_1(n_1+1)}{2}.$$

An exact p-value is computed when both groups contain fewer than 50 observations and the data contain no ties; otherwise a normal approximation with continuity correction is used.

9.5 Fisher's one-way ANOVA and Welch's heteroscedastic ANOVA

Fisher's ANOVA test statistic (Fisher and Yates, 1990) is

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(N-k)} = \frac{\sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2 \ / \ (k-1)}{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2 \ / \ (N-k)},$$

where MS_{between} and MS_{within} are the between-group and within-group mean squares, SS_{between} and SS_{within} are the corresponding sums of squares, k is the number of groups, $N = \sum_{i=1}^{k} n_i$ is the total sample size, $\bar{x}_i$ is the mean of group i, $\bar{x}$ is the overall mean, and x_{ij} is observation j in group i. Under the null hypothesis $F \sim F(k-1, N-k)$. Post-hoc comparisons use `TukeyHSD()` (Hochberg and Tamhane, 1987), which controls the family-wise error rate via the studentised range distribution (Abdi, 2007).

Welch's heteroscedastic ANOVA (`oneway.test()`) (Welch, 1951) down-weights groups with large variance. Its test statistic is

$$F_W = \frac{\sum_{i=1}^{k} w_i(\bar{y}_i - \bar{y}_w)^2 \ / \ (k-1)}{1 + \frac{2(k-2)}{k^2-1}\sum_{i=1}^{k}\frac{(1-w_i/w)^2}{n_i - 1}},$$

where $w_i = n_i/s_i^2$ are the inverse-variance weights, $w = \sum_{i=1}^{k} w_i$, and $\bar{y}_w = \sum_{i=1}^{k} w_i\bar{y}_i/w$ is the weighted grand mean. Degrees of freedom are adjusted via a Satterthwaite-type approximation (Satterthwaite, 1946). Post-hoc comparisons use `games.howell()` (Games and Howell, 1976), which applies separate variance estimates per pair. Compact letter displays are produced by `multcompLetters()` from **multcompView** (Graves et al., 2026).

9.6 Kruskal–Wallis test

When normality is rejected, `kruskal.test()` (Kruskal and Wallis, 1952) tests whether the groups come from the same population based on ranks:

$$H = \frac{12}{N(N+1)}\sum_{i=1}^{k} n_i\left(\bar{R}_i - \bar{R}\right)^2,$$

where n_i is the sample size of group i, k is the number of groups, $\bar{R}_i$ is the average rank of group i, N is the total sample size, and $\bar{R} = (N+1)/2$ is the expected average rank under the null hypothesis. The statistic approximately follows $\chi^2(k-1)$. Post-hoc comparisons use Holm-adjusted `pairwise.wilcox.test()` (Holm, 1979).

9.7 Pearson's χ^2 test and Fisher's exact test

Let O_{ij} and E_{ij} denote the observed and expected frequencies in row i and column j of an $R \times C$ contingency table, where rows index the R levels of the response y and columns the C levels of the predictor x. The Pearson residual for cell (i, j) is

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}, \quad i = 1, \ldots, R, \quad j = 1, \ldots, C.$$

The test statistic of Pearson's χ^2 test (Pearson, 1900) is

$$\chi^2 = \sum_{i=1}^{R} \sum_{j=1}^{C} r_{ij}^2 = \sum_{i=1}^{R} \sum_{j=1}^{C} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

compared to $\chi^2((R-1)(C-1))$. For 2×2 tables, Yates' continuity correction (Yates, 1934) is applied by default. For general $R \times C$ tables, `visstat()` supplements the bar chart with a mosaic plot in which tiles are coloured by r_{ij} (blue: positive, red: negative).

Fisher's exact test (Fisher, 1970) is applied when Cochran's rule (Cochran, 1954) is violated. The test calculates an exact p-value by conditioning on the observed margins of an $R \times C$ contingency table. Let $\mathcal{T} = (n_{ij})$ denote the observed table. To maintain consistency with the `y ~ x` (response ~ predictor) framework used throughout `visStatistics`, rows ($i = 1, \ldots, R$) represent the levels of the response variable y and columns ($j = 1, \ldots, C$) represent the levels of the predictor x. The row totals are $n_{i\cdot} = \sum_{j=1}^{C} n_{ij}$ and the column totals are $n_{\cdot j} = \sum_{i=1}^{R} n_{ij}$.

In the 2×2 case ($R = 2, C = 2$), the table is structured as follows:

	x_1	x_2	Row sums
y_1	n_{11}	n_{12}	$n_{1\cdot}$
y_2	n_{21}	n_{22}	$n_{2\cdot}$
Column sums	$n_{\cdot 1}$	$n_{\cdot 2}$	n

The exact probability of observing this table under the null hypothesis of independence, given the fixed margins, is given by the hypergeometric probability mass function:

$$\mathbb{P}(\mathcal{T} \mid n_{1\cdot}, n_{2\cdot}, n_{\cdot 1}, n_{\cdot 2}) = \frac{\binom{n_{1\cdot}}{n_{11}}\binom{n_{2\cdot}}{n_{21}}}{\binom{n}{n_{\cdot 1}}},$$

where $n = n_{1\cdot} + n_{2\cdot} = n_{\cdot 1} + n_{\cdot 2}$ is the total sample size. The p-value is computed by summing the probabilities of all tables with the same margins whose probabilities under the null are less than or equal to that of the observed table. For general $R \times C$ tables, `fisher.test()` generalises this approach using the multivariate hypergeometric distribution. For 2×2 tables, `fisher.test()` additionally returns the conditional maximum likelihood estimate of the odds ratio $\widehat{\mathrm{OR}} = n_{11}n_{22}/(n_{12}n_{21})$ and its confidence interval.

9.8 Rank Association Measures

Spearman rank correlation

For two numeric variables with `correlation = TRUE`, `visstat()` calls `cor.test(x, y, method = "spearman")` to measure the monotonic association between x and y using ranks, evaluating linear dependence on the ranked data rather than on the original scale. Spearman's ρ is Pearson's r applied to the ranks:

$$\rho = r(\mathrm{rank}(x), \mathrm{rank}(y))$$

where $r(u, v)$ denotes Pearson's correlation coefficient:

$$r(u, v) = \frac{\sum_{i=1}^{n}(u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^{n}(u_i - \bar{u})^2}\sqrt{\sum_{i=1}^{n}(v_i - \bar{v})^2}}.$$

Here $u_i = \mathrm{rank}(x_i)$ and $v_i = \mathrm{rank}(y_i)$ are the ranks of the n paired observations, and $\bar{u}$ and $\bar{v}$ are their sample means.

For inference, `cor.test(..., method = "spearman")` computes an exact p-value for small samples without ties by evaluating all $n!$ rank permutations. For larger samples or when ties are present, it uses an approximation to the null distribution of the rank association measure or its asymptotic transformation. No distributional assumptions on the original data are required.

Kendall's τ_b

For two ordinal variables with n joint observations, let C denote the number of concordant pairs (those whose ranks agree in both variables) and D the number of discordant pairs. Kendall's τ_b is defined as

$$\tau_b = \frac{C - D}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}$$

where $n_0 = n(n-1)/2$, $n_1 = \sum_i t_i(t_i - 1)/2$ with t_i denoting the sizes of tie groups in the response, and $n_2 = \sum_j u_j(u_j - 1)/2$ with u_j denoting the sizes of tie groups in the predictor. The denominator correction makes τ_b attain ± 1 even with ties, which Spearman's ρ does not (Kendall, 1945). With few ordered levels (e.g., five-point Likert items), ties are unavoidable; this is the principal reason to prefer τ_b over Spearman's ρ in this setting (Agresti, 2010).

`visstat()` calls `cor.test(as.numeric(y), as.numeric(x), method = "kendall", exact = FALSE)` and reports τ_b, the asymptotic test statistic $z = \tau_b / \mathrm{SE}(\tau_b)$, and the two-sided p-value.

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